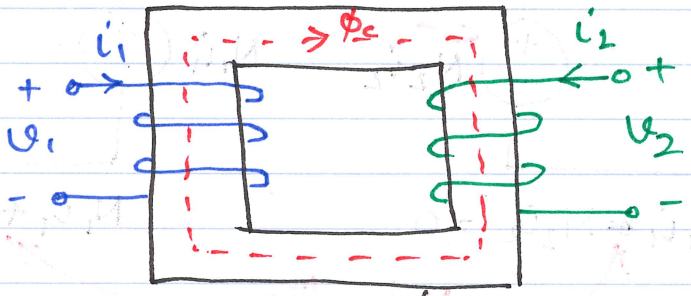
Lecture 11

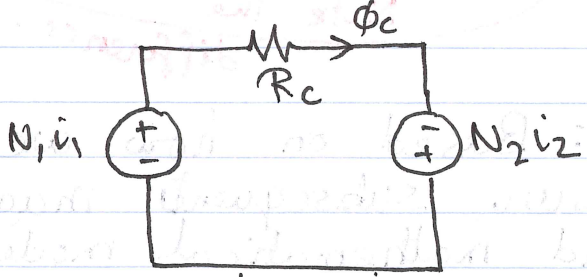
Last time : (1) Flux density computation and why ?
(2) Flux linkage – "Why is pending"
(3) Coupled coils . – Eg: Transformer .
Wireless power charger .

Today : (1) Coupled coil → dot convention
→ complete the model . → Mutual inductance .

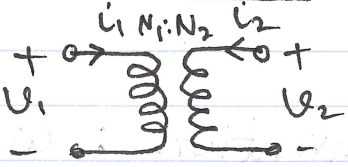
(1) We started with the assumption that the core is infinitely permeable .



Physical Model .



Magnetic Equivalent Circuit .



Electric ~~equivalent~~ circuit model .

Assuming $\mu_r \to \infty$, $R_c \to 0$ ∴ $\boxed{N_1 i_1 + N_2 i_2 = 0}$.

$$v_1 = \frac{d\lambda_1}{dt} = N_1 \frac{d\phi_c}{dt}$$

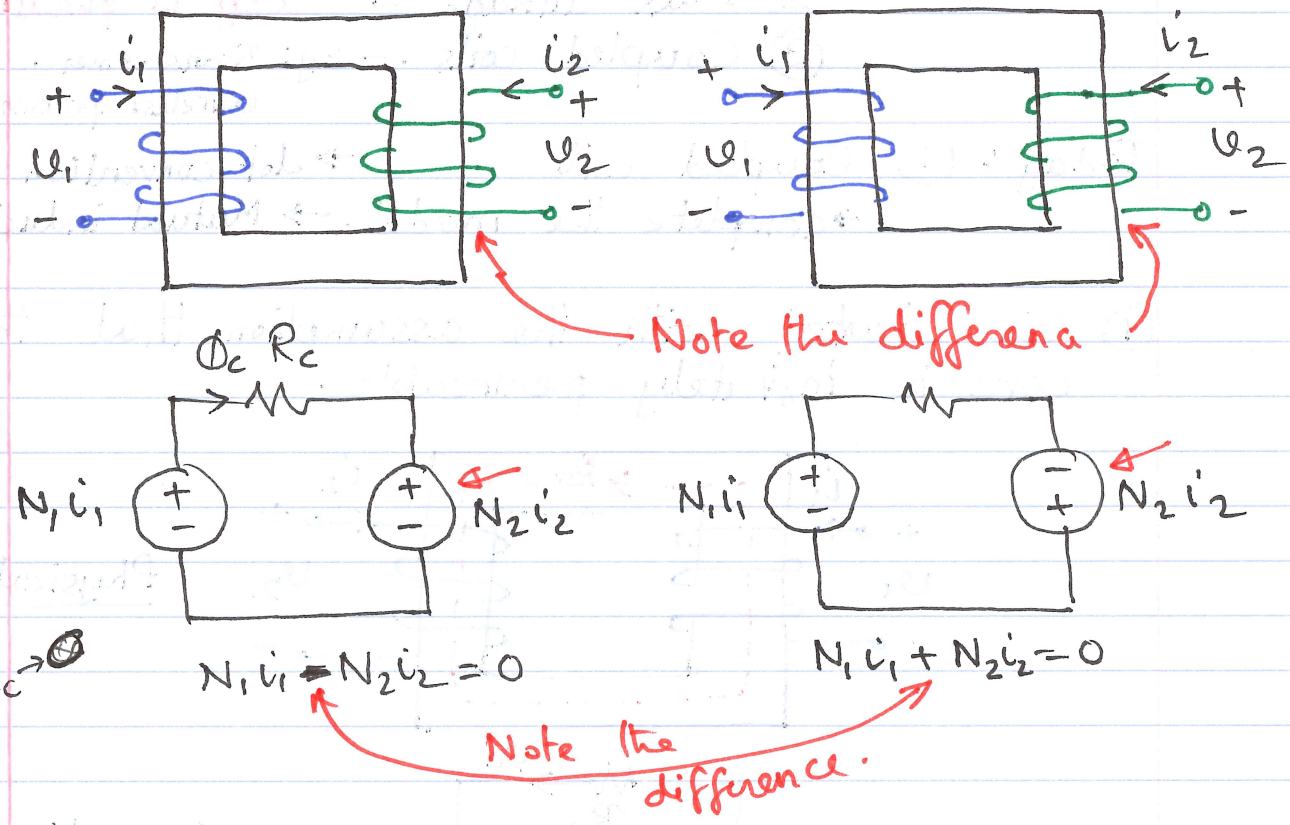
$$v_2 = \frac{d\lambda_2}{dt} = N_2 \frac{d\phi_c}{dt}$$

$$\boxed{\frac{v_1}{N_1} = \frac{v_2}{N_2}}$$

Mathematical model

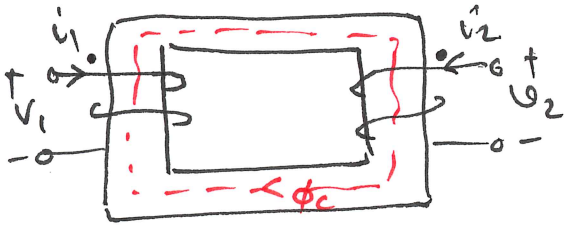
in the physical model.

We have one problem: What happens if we "wound" coils in a different way:



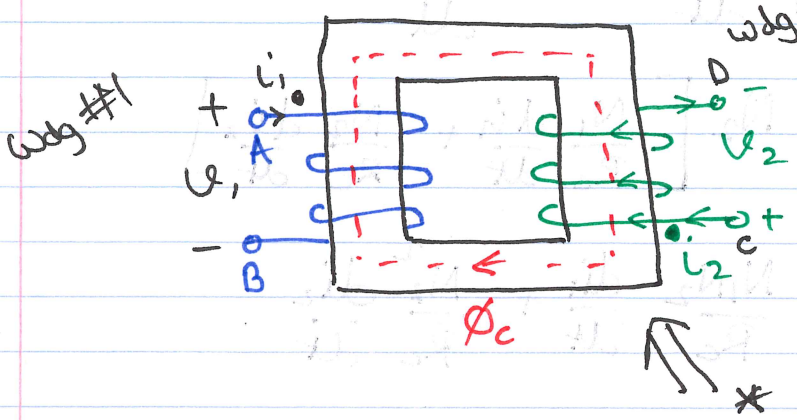
Bottomline: Based on how we wound physically, changes our subsequent magnetic equivalent circuit and mathematical model. Kind of annoying !!

Fix: Dot convention: We will place "dots" on the windings such that we set our reference voltage and currents follow the same mathematical model as well as magnetic equivalent circuit model.



⟸ * Both have same.
MEC and $\frac{v_1}{v_2} = \frac{N_1}{N_2}$
$N_1 i_1 + N_2 i_2 = 0$.

Dot convention: "Current entering the dots throws flux in the same direction".



① Place a dot in one winding arbitrarily (Eg: #1)
② Mark that terminal (Eg: A) with positive voltage polarity and current enters the dot.
③ Draw the flux path ϕ_c.
④ Determine current in wdg #2 which will throw flux in the same direction as the original wdg.
⑤ Put the "dot" in the second wdg, where current enters the terminal (Eg: C).
⑥ Put the polarity of voltage V_2 as (+) in the same terminal.

Next, let us throw away the assumption that the core is infinitely permeable.

$$N_1 i_1 + N_2 i_2 = \phi_c R_c \neq 0 .$$

$$\therefore \quad V_1 = \frac{d\lambda_1}{dt} = N_1 \frac{d\phi_c}{dt} = N_1 \frac{d}{dt}\left(\frac{N_1 i_1 + N_2 i_2}{R_c}\right)$$

$$= \frac{N_1^2}{R_c}\frac{di_1}{dt} + \frac{N_1 N_2}{R_c}\frac{di_2}{dt}$$

$$v_2 = \frac{d\lambda_2}{dt} = N_2 \frac{d\phi_c}{dt}$$

$$= N_2 \left[\frac{N_1}{R_c} \frac{di_1}{dt} + \frac{N_2}{R_c} \frac{di_2}{dt} \right]$$

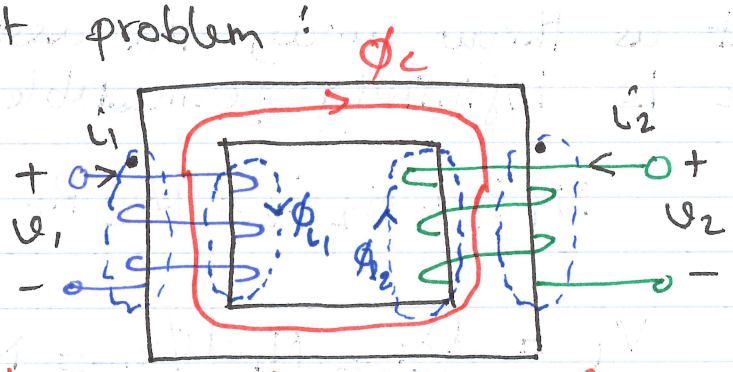
$$= \frac{N_1 N_2}{R_c} \frac{di_1}{dt} + \frac{N_2^2}{R_c} \frac{di_2}{dt}$$

Recall: $\frac{N_1^2}{R_c}$: Inductance (Self inductance for wdg 1)

$\frac{N_2^2}{R_c}$: Self inductance of wdg 2

$\frac{N_1 N_2}{R_c}$: Mutual inductance of coil 1 and coil 2.

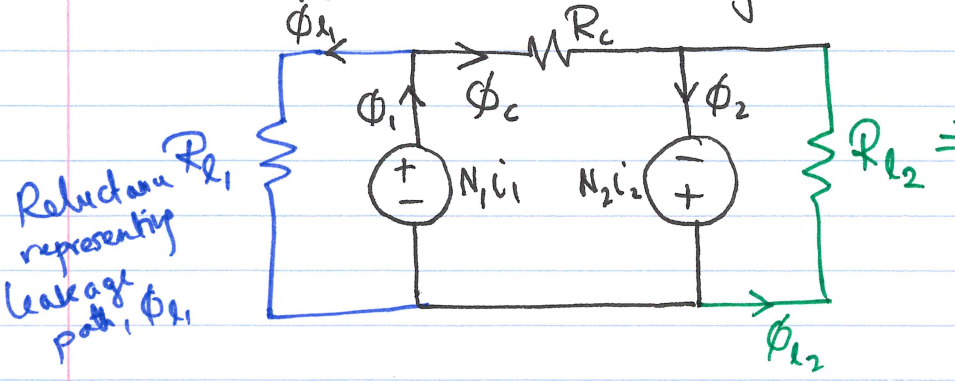
One last problem:



What happens if all the flux does not flow through the core?

Very much possible because μ_r is finite ϕ_{l_1}, ϕ_{l_2}: Leakage flux

Let us redraw the magnetic equivalent circuit.



Reluctance representing leakage path, ϕ_{l_1}

$\Rightarrow$ Reluctance representing leakage path, ϕ_{l_2}.

$$\therefore \quad \phi_1 = \phi_{l_1} + \phi_c$$

$$= \frac{N_1 i_1}{R_{l_1}} + \frac{N_1 i_1 + N_2 i_2}{R_c}$$

$$\therefore \quad \lambda_1 = N_1 \phi_1 = \left(\frac{N_1^2}{R_{l_1}} + \frac{N_1^2}{R_c}\right) i_1 + \left(\frac{N_1 N_2}{R_c}\right) i_2$$

$$\therefore \quad V_1 = \frac{d\lambda_1}{dt} = \left(\frac{N_1^2}{R_{l_1}} + \frac{N_1^2}{R_c}\right) \frac{di_1}{dt} + \left(\frac{N_1 N_2}{R_c}\right) \frac{di_2}{dt}$$

Similarly, $V_2 = \frac{d\lambda_2}{dt} = \left(\frac{N_1 N_2}{R_c}\right) \frac{di_1}{dt} + \left(\frac{N_1^2}{R_{l_2}} + \frac{N_1^2}{R_c}\right) \frac{di_2}{dt}$

Self-inductance of wdg #1 $= \left(\frac{N_1^2}{R_c} + \frac{N_1^2}{R_{l_1}}\right) = L_{11}$

" of " #2 $= \left(\frac{N_2^2}{R_c} + \frac{N_2^2}{R_{l_2}}\right) = L_{22}$

Mutual inductance, $= \frac{N_1 N_2}{R_c} = M$.

$$\therefore \quad \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} L_{11} & M \\ M & L_{22} \end{bmatrix} \begin{bmatrix} di_1/dt \\ di_2/dt \end{bmatrix}.$$

